EA Assistant — Product Requirements Document

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1. Product Overview

EA Assistant is a Claude Code plugin for **EA Practitioners** leading Enterprise Architecture engagements. It provides a structured, interview-driven workflow grounded in **TOGAF 10** (process), **Zachman Framework** (classification), and **ArchiMate 3.x** (notation).

The plugin turns Claude into an EA facilitator: it interviews practitioners and stakeholders, populates TOGAF artifacts from those interviews, tracks decisions and traceability chains, and produces consolidated architecture reports. It replaces the unstructured Word/spreadsheet/shared-drive workflow that most teams use to run engagements today.

It is not an EA modelling tool (that is Sparx, Archi, or MEGA). It is the **engagement management and documentation layer** that most EA tooling omits.

Platform: The current version runs inside **Claude Code** and relies on the Claude Code plugin framework (commands, skills, agents). A future version is planned as a standalone application backed by an LLM API, removing the Claude Code dependency. Implementation decisions should avoid deep Claude Code-specific coupling where a framework-agnostic alternative exists.

2. Problem Statement

Enterprise Architecture engagements typically span 3–6 months, involve 5–15 stakeholders, and produce 10–20 artifacts across 4–8 ADM phases. Practitioners face three core problems:

- **Structure without tools** — TOGAF provides a framework but no execution tooling. Teams manage engagements in Word documents, spreadsheets, and shared drives with no traceability or consistency enforcement. Artifacts drift out of sync; decisions are undocumented; there is no single source of truth for engagement state.
- **Interview quality** — Capturing good architecture direction (drivers, goals, objectives) requires skilled facilitation. Concepts like Goal vs. Objective vs. Strategy are routinely confused, producing artifacts whose motivation chain cannot be traced and whose objectives cannot be measured. A poorly facilitated Phase A interview produces an Architecture Vision that cannot drive downstream phases.
- **Continuity** — EA engagements span multiple sessions and facilitators. Context is lost between sessions, decisions made in one session are not visible in the next, and artifacts populated by different people contradict each other.

3. Target Users

The **EA Practitioner** is the primary user. All design decisions should favour this user when trade-offs arise.

Priority	User	Role	Primary Need	Entry Point
1	**EA Practitioner**	Lead architect running the engagement	Structured workflow, artifact generation, traceability	`/ea-new` , `/ea-open` , `/ea-phase`
2	**EA Facilitator**	Runs stakeholder interviews (may be same person as practitioner)	Question bank, guided interview flow, session notes	`/ea-interview` , `/ea-brainstorm`
3	**Sponsor**	Authorises the engagement; provides strategic direction	Consolidated reports, Decision Register filtered to their level	`/ea-publish` , `/ea-decisions --audience executive`
4	**Business Analyst**	Captures and manages architecture requirements	Requirements Register with motivation traceability	`/ea-requirements`

> Stakeholders (non-sponsor) participate in interviews but do not use the tool directly — they interact via the Web or Voice interview form.

4. Core Concepts

Motivation Framework

The engagement's strategic context is captured as a linked chain:

```
Business Drivers (DRV) ──drives──► Goals (G) ──operationalises──► Objectives (OBJ)
                                        ▲                                ▲
                              threatens │                         blocks │
                                        │                                │
                                   Issues (ISS)                   Problems (PRB)
                                                                        │
                                                              Requirements Register
                                                             (links to any of the above)
```

- **Business Drivers** — forces making the engagement necessary (internal/external, opportunity/threat/mandate)
- **Goals** — qualitative desired outcomes linked to drivers

- **Objectives** — measurable, time-bound results that operationalise goals
- **Issues** — systemic barriers that *threaten* goals (not observable symptoms — structural, persistent)
- **Problems** — specific, observable symptoms that *block* objectives
- **Strategies** — chosen approaches for achieving goals; recorded in §7 Strategic Direction Summary of the Architecture Vision (STR-NNN)

Requirements Register entries carry a Motivation field that links each requirement to its source — any of: DRV, ISS, PRB, G, or OBJ.

EA Concepts (8 total)

Principle, Goal, Objective, Strategy, Plan, Risk, Issue, Problem — each with a formal definition, TOGAF phase placement, ArchiMate 3.x element, and a disambiguation checklist to prevent concept confusion during interviews. Full definitions in `skills/ea-artifact-templates/references/ea-concepts.md` .

Disambiguation summary:

Concept	Qualitative or Measurable	Time-bound	Owns a mitigation	Links to
Principle	Rule (non-negotiable)	No	No	Architecture decisions
Goal	Qualitative	No	No	Drivers
Objective	Measurable	Yes	No	Goals
Strategy	Directional	No	No	Goals (STR-NNN)
Plan	Ordered action set	Yes	No	Strategies
Issue	Qualitative (systemic barrier)	No	Yes (action plan)	Goals (threatens)
Problem	Specific symptom	No	Yes (requirement)	Objectives (blocks)
Risk	Potential future event	No	Yes (mitigation)	Goals or Objectives

5. Features

5.1 Engagement Management

- Create engagements with type (Greenfield / Brownfield / Assessment-only / Migration), domains, sponsor, scope, and dates
- Open, edit metadata, archive, restore, delete
- Portfolio dashboard (`/ea-status`) showing all engagements with: type, domains, current phase, artifact counts, opt-out count, non-standard artifact flags
- Per-engagement `CLAUDE.md` auto-generated on `/ea-new` ; refreshed (overwritten from template + current engagement state) on every `/ea-open` — loads session context automatically when Claude is opened from the engagement folder

> Note: The `CLAUDE.md` refresh on `/ea-open` overwrites the file. Do not add manual notes to `CLAUDE.md` ; use `brainstorm/brainstorm-notes.md` instead.

5.2 ADM Lifecycle

- 11 phases: Prelim, Requirements (custom non-TOGAF phase for structured requirements capture), A–H
- Phase status tracking with timestamps (Not Started / In Progress / Complete / On Hold / Not Applicable)
- Non-linear navigation — jump to any phase at any time
- **Phase applicability** is determined by engagement type and domain selection at creation time:

Phase	Greenfield	Brownfield	Assessment-only	Migration
Prelim	Required	Required	Required	Required
Requirements	Required	Required	Required	Required
A — Architecture Vision	Required	Required	Required	Required
B — Business Architecture	Required	Required	Optional	Required
C — Data Architecture	Domain†	Domain†	Optional	Domain†
C — Application Architecture	Domain†	Domain†	Optional	Domain†
D — Technology Architecture	Domain†	Domain†	Optional	Domain†
E — Opportunities & Solutions	Required	Required	Not Applicable	Required
F — Migration Planning	Optional	Required	Not Applicable	Required
G — Implementation Governance	Required	Required	Not Applicable	Required
H — Architecture Change Mgmt	Optional	Required	Not Applicable	Optional

† *Domain-dependent*: phase is Required if the corresponding domain (Data / Application / Technology) was selected; Not Applicable if not selected.

5.3 Phase Interviews

- Curated question banks for every ADM phase with output routing tables linking each question to its target artifact section
- Four modes:
  - **Web** (default) — interactive input form rendered as a React artifact; one question card at a time with progress bar
  - **Voice** — Web mode with 🎤 mic button per question; uses `window.SpeechRecognition` ; transcript is editable before submission; falls back gracefully if speech recognition unavailable
  - **Text** — chat-based Q&A; question preview with 6-option menu before answering
  - **Display** — read-only list of all questions for the phase/artifact; no answer recording; used for preparation or reference
- **Question preview** (Text mode) — shows full question list before starting; menu options: Start Web / Voice / Text, Brainstorm first, Jump to specific question, Display only, Resume (if prior answers exist)

- **ID scheme for Phase A:** DRV-NNN, G-NNN, OBJ-NNN, ISS-NNN, PRB-NNN, STR-NNN, MET-NNN — documented in the phase header; each question group carries a `§N` section marker linking to its target Architecture Vision section
- Session attribution — step 0 of every interview collects facilitator name and participant list; recorded in session log and interview note frontmatter
- Chronological session log per engagement (`interviews/session-log.md`); prior session summary shown at the start of each new session

5.4 Artifact Generation

Artifacts are populated from interview answers, uploaded documents, or explicit user input. No AI-generated content is written to an artifact without a 🤖 `AI Draft — Review Required` marker.

15 TOGAF artifact templates:

Artifact	Phase	A3 Decision Log
Architecture Principles	Prelim	—
Requirements Register (with Motivation field)	Requirements	—
Traceability Matrix	Requirements	—
Architecture Vision (15 sections)	A	✓
Statement of Architecture Work	A	✓
Stakeholder Map	A	—
Business Architecture	B	✓
Business Model Canvas	B	—
Data Architecture	C-Data	—
Application Architecture	C-App	—
Technology Architecture	D	—
Gap Analysis (covers all selected domains)	B–D	—
Architecture Roadmap	E	✓
Migration Plan	F	—
Architecture Contract	G	✓
Compliance Assessment	G	—

Architecture Vision sections:
§1 Executive Summary · §2 Business Drivers · §3 Goals · §4 Objectives · §5 Issues · §6 Problems · §7 Strategic Direction Summary · §8 Scope · §9 Stakeholders · §10 Architecture Principles · §11 Constraints · §12 Assumptions · §13 High-Level Target Architecture · §14 Key Risks · §15 Next Steps · Appendix A3 Decision Log

Answer state markers:

- ⚠️ `Not answered` — skipped, can return later
- ➖ `Not applicable`
- ⊘ `Opted out – {reason}` — deliberate exclusion, tracked
- 🤖 `AI Draft – Review Required`
- ✓ `Default accepted`
- 📎 `Source: uploads/{file}`

5.5 Interview Quality Controls

- **Compliance check** — see §9 Quality Gates
- **Cross-topic detection** — flags answers that belong in a different artifact; offers to route immediately, flag for later, or continue as-is
- **Concept-check** — catches Goal/Objective/Strategy/Issue/Problem confusion inline; offers reclassification with reference to the EA Concepts disambiguation table
- **Opt-out tracking** — question and artifact level; reason + timestamp recorded in `engagement.json → optOuts[]`; surfaced in `/ea-status` and consolidated reports
- **Brainstorm integration** — freeform thoughts captured via `/ea-brainstorm` are matched to interview questions by keyword overlap and surfaced as 💭 hints when a question's subject matches the brainstorm content

5.6 Facilitator Config

Controlled via `.claude/ea-assistant.local.md` in the working directory:

Setting	Options	Default	Effect
`facilitatorStyle`	`patient` / `direct` / `executive`	`patient`	Tone, pacing, acknowledgement, section pauses
`audienceLevel`	`executive` / `architect` / `technical` / `mixed`	`mixed`	Terminology depth and TOGAF jargon level
`requireConfirmBeforeRecord`	`true` / `false`	`false`	Confirm before writing each answer to artifact
`researchPrompts`	`true` / `false`	`true`	Show `@research-agent` prompts on drivers/risks/assumptions
`sessionSummary`	`true` / `false`	`true`	End-of-session topic/theme summary

Style behaviour:

	`patient`	`direct`	`executive`
Preamble	One sentence why question matters	Question only	Business-outcome framing
Acknowledgement	Brief and warm	None	None
Short answer	One gentle probe	Accept as-is	Accept as-is

	`patient`	`direct`	`executive`
Examples	Proactive	On request	On request
Transitions	"Anything else?" pause	None	Checkpoint every 5–7 Qs
Jargon	TOGAF with gloss	Full TOGAF	Business language only

Precedence: When `facilitatorStyle` and `audienceLevel` conflict (e.g., `executive` style with `technical` audience), `audienceLevel` governs terminology depth and `facilitatorStyle` governs pacing and tone. The practitioner is responsible for setting a coherent combination.

5.7 Decision Register

Decisions captured during interviews are logged in **Appendix A3 Decision Log** tables, which are included in the five key artifacts marked ✓ in §5.4. Other artifacts may include A3 optionally.

`/ea-decisions` aggregates all A3 rows across artifacts into a cross-artifact register.

Governance states and semantics:

State	Meaning	Who sets it
`Provisional`	Decision recorded but not yet reviewed	Interviewer (automatic on A3 log)
`Awaiting`	Under review; stakeholder confirmation or governance vote pending	EA Practitioner
`Verified`	Confirmed correct by the responsible architect; no formal vote required	EA Practitioner
`Voted`	Approved by a formal governance body or quorum	Sponsor / governance body
`Fiat`	Accepted by executive authority without formal vote; sponsor directive	Sponsor
`Returned`	Sent back for rework; reason must be recorded	Reviewer

Transition path: Provisional may move directly to Verified, Voted, or Fiat when no formal review step is needed. Awaiting is an optional holding state used when stakeholder confirmation or a governance vote is pending before resolution. Returned sends the decision back to Provisional; a reason must be recorded.

```
Provisional ──────────────────────────────► Verified / Voted / Fiat
     │                                              ▲
     └──► Awaiting (optional) ──────────────────────┘
                                  │
                                  └──► Returned ──► Provisional
```

Filter flags: `--audience` , `--owner` , `--domain` , `--authority` , `--cost` , `--impact` , `--risk` , `--subject` , `--status`
Audience presets: `executive` / `architect` / `business` / `technical`

5.8 Artifact Review and Grill

Review workflow:

- States: Not Reviewed → In Review → Approved / Needs Revision → (if Needs Revision) back to In Review
- `/ea-review [artifact]` — opens the artifact, runs the compliance check, and presents the review interface where the reviewer can update status, add a review note, and trigger Needs Revision with a reason
- Approved artifacts cannot be overwritten without confirmation; the interviewer is warned before writing to an Approved field

Grill (deep review):

- `/ea-grill [artifact] [--skill]` — runs a grill-me skill against the artifact; output saved to `reviews/{artifact}-review-{YYYY-MM-DD}.md`
- Auto-selects skill by artifact type; override with `--skill`
- When an artifact matches multiple routing rows, the more specific row wins (e.g., Architecture Vision matches both "Architecture Vision, Strategy" and "Any structured document" — `stress-test` wins)

Grill skill routing (auto-selection):

Artifact type	Default skill
Architecture Vision, Strategy	stress-test
Architecture Roadmap, Migration Plan	premortem
Architecture Principles, Decisions	decision
Business Architecture, BMC	design
Application Architecture	software-design
Technology Architecture, Infra	infra-design
Any structured document	artifact
Any diagram	diagram
Executive presentation	boardroom-strategy

5.9 Publishing

- **Pre-publish compliance check** — all selected artifacts scanned before assembly; non-compliant items flagged with option to proceed or remediate
- Consolidates artifacts in TOGAF ADM order (Prelim → Requirements → A → B → C-Data → C-App → D → E → F → G → H) into a single Markdown or Word document
- Cover page, Artifact Status Summary table, Table of Contents, per-artifact status headers
- Opted-out items and non-standard artifacts flagged inline

Export tools:

- `/ea-publish` — uses **Pandoc** to render Markdown → `.docx`; requires `pandoc` installed

- `/ea-generate [artifact] word` — uses **python-docx** for per-artifact Word export; requires `python-docx`
- `/ea-generate [artifact] pptx` — uses **python-pptx** for PowerPoint export; requires `python-pptx`
- `/ea-generate [artifact] mermaid` — renders Mermaid diagram source from artifact content

5.10 Research Agent Integration

- Invoke `@research-agent` at any interview prompt or during facilitation by typing `@research-agent` followed by the claim to validate
- Validates business drivers, technology claims, risks, and assumptions with cited evidence
- If no supporting evidence is found, the agent states this explicitly — absence of evidence is surfaced, not silently omitted
- Findings are pasted by the user directly into the current answer field or brainstorm notes; they are not auto-written to artifacts
- Research prompts are shown automatically on driver, risk, and assumption questions when `researchPrompts: true` (default)
- Documented in Phase A facilitation notes and `/ea-help`

5.11 Undocumented Agents (Planned Features)

Three agents exist in the plugin but have no dedicated command workflow yet. They are loaded as part of the plugin context and invoked by asking Claude directly:

- **`ea-consistency-checker`** — flags cross-artifact inconsistencies (e.g., a Goal in the Architecture Vision that has no corresponding entry in the Business Architecture); invoke by asking: *"Check for cross-artifact inconsistencies in this engagement."*
- **`ea-document-analyst`** — analyses uploaded documents in `uploads/` for architecture content and suggests artifact mappings; invoke by asking: *"Analyse the uploaded documents and suggest artifact mappings."*
- **`ea-advisor`** — answers EA methodology questions (TOGAF, Zachman, ArchiMate) in context; invoke by asking any methodology question in chat.

These will gain dedicated commands and workflow integration in a future version.

6. Data Model

Folder Structure

```
EA-projects/
└── {slug}/
    ├── engagement.json        # all state: phases, artifacts, sessions, direction, metrics, optOuts
    ├── CLAUDE.md              # auto-generated session context; refreshed (overwritten) on /ea-open
    ├── artifacts/             # .md artifact files + .review.md review files
    ├── interviews/            # session-log.md + dated interview notes per session
    ├── brainstorm/            # brainstorm-notes.md
    ├── diagrams/              # .mmd, .dot, .drawio files
    ├── uploads/               # source documents for ingestion
    ├── reviews/               # grill-me review outputs
    └── ui/                    # generated HTML interview/brainstorm forms
```

engagement.json Schema

```
{
  "name": "",
  "slug": "",
  "description": "",
  "sponsor": "",
  "organisation": "",
  "scope": "",
  "engagementType": "Greenfield | Brownfield | Assessment-only | Migration",
  "architectureDomains": ["Business", "Data", "Application", "Technology"],
  "startDate": "YYYY-MM-DD",
  "targetEndDate": "YYYY-MM-DD or null",
  "status": "Active | On Hold | Planning | Completed",
  "currentPhase": "Prelim",
  "requirementsRepoPath": "",
  "lastModified": "",
  "direction": {
    "Business": {
      "goals": [{ "id": "G-001", "statement": "", "priority": "" }],
      "objectives": [{ "id": "OBJ-001", "statement": "", "measure": "", "target": "", "deadline": "", "priority":
      "strategies": [{ "id": "STR-001", "statement": "", "supports": ["G-001"], "priority": "" }]
    }
  },
  "metrics": {
    "Business": [{ "id": "MET-001", "name": "", "type": "outcome | performance | activity", "measure": "", "basel
  },
  "phases": {
    "Prelim": { "status": "Not Started", "startedAt": null, "completedAt": null }
  },
  "sessions": [
    {
      "id": "session-001",
      "date": "YYYY-MM-DD",
      "facilitator": "",
      "participants": [""],
      "phase": "",
      "artifactsWorked": [""],
      "topics": [""],
      "nextStep": "",
      "notesFile": "interviews/interview-YYYY-MM-DD.md"
    }
  ],
  "artifacts": [{ "id": "", "name": "", "phase": "", "file": "", "status": "Draft", "reviewStatus": "Not Reviewec
  "optOuts": [{ "type": "question | artifact", "artifactId": "", "questionRef": "", "reason": "", "timestamp": ""
}
```

> `direction` and `metrics` keys are domain-scoped. When multiple domains are active (e.g., Business + Technology), each domain gets its own key at the same level as `"Business"` in the example above.

7. Commands

Command	Key Arguments / Options	Description
`/ea-new`	—	Create engagement — collects name, type, domains, sponsor, scope, dates; generates CLAUDE.md
`/ea-open`	`[slug]`	Open engagement, refresh CLAUDE.md, next-action menu
`/ea-status`	—	Portfolio dashboard — all engagements with progress, opt-outs, non-standard flags
`/ea-phase`	`[phase name]`	Start, navigate to, or resume an ADM phase
`/ea-interview`	`[web	voice
`/ea-artifact`	`[create	view
`/ea-brainstorm`	`[phase]`	Capture freeform thoughts before or during interviews

Command	Key Arguments / Options	Description
`/ea-generate`	`[artifact] [word	pptx
`/ea-review`	`[artifact]`	Open artifact for review; runs compliance check; update review status
`/ea-grill`	`[artifact] [--skill name]`	Deep-review artifact using a grill-me skill; auto-selects skill by type
`/ea-requirements`	`[list	add
`/ea-decisions`	`[--audience] [--owner] [--domain] [--status] [--cost] [--impact] [--risk]`	Generate Decision Register from all A3 logs with filters
`/ea-publish`	`[markdown	word]`
`/ea-help`	—	Command reference, interview shortcuts, research agent guide

8. Agents

Agent	Role	Invoked by
`ea-facilitator`	Guides users through ADM phases; reads facilitatorStyle config	`/ea-phase` , `/ea-open`
`ea-interviewer`	Conducts structured interviews; all 4 modes, question preview, brainstorm, cross-topic detection	`/ea-interview`

Agent	Role	Invoked by
`ea-requirements-analyst`	Extracts structured requirements from uploaded documents	`/ea-requirements`
`ea-consistency-checker`	Flags cross-artifact inconsistencies (no dedicated command)	Ask Claude: "Check for cross-artifact inconsistencies"
`ea-document-analyst`	Analyses uploaded documents for architecture content (no dedicated command)	Ask Claude: "Analyse the uploaded documents"
`ea-advisor`	Answers EA methodology questions — TOGAF, Zachman, ArchiMate (no dedicated command)	Ask any methodology question in chat
`ea-diagram`	Generates and interprets architecture diagrams	`/ea-generate [artifact] mermaid`

9. Quality Gates

Gate	When	Mechanism
Artifact compliance	Every artifact open (interview, review, view)	3-tier check: frontmatter → template structure → artifact-specific requirements (e.g., A3 presence in key artifacts)
Pre-publish compliance	Before `/ea-publish` assembles	All selected artifacts scanned; user chooses proceed or remediate; non-compliant items flagged in output
Content policy	Throughout interview	No AI content without 🤖 marker; no overwrite of Approved fields without confirmation
Opt-out audit	Ongoing	Every exclusion tracked with reason + timestamp in `engagement.json → optOuts[]`; surfaced in `/ea-status` and published reports

10. Success Metrics

Metric	Baseline (without tool)	Target	How measured
Phase A interview duration (experienced EA)	~3 hours across 2–3 sessions	≤ 90 minutes in a single session	Session log timestamps from start of Phase A interview to

Metric	Baseline (without tool)	Target	How measured
			Architecture Vision Draft status
Motivation chain coverage	Typically 0–20% of objectives have a traceable driver	100% of OBJ-NNN entries link to at least one G-NNN; 100% of G-NNN entries link to at least one DRV-NNN	Compliance check on Architecture Vision §2–§6
Report manual corrections before stakeholder submission	Typically 2–4 hours of formatting and gap-filling	≤ 3 corrections (missing fields, formatting) per published report	User-reported on post-publish review
Non-EA stakeholder participation rate	Low — stakeholders disengage from text-heavy TOGAF interviews	Non-EA stakeholders can complete a Web or Voice interview with no prior TOGAF knowledge, verified by facilitator observation	Facilitator assessment at session close
Decision traceability	Decisions made verbally in sessions; rarely documented	All A3 log entries have a recorded owner and governance state of Awaiting or above before Phase G commences	`/ea-decisions` output inspected at Phase G gate: zero entries with state `Provisional` and no `owner` field

11. Out of Scope (Current Version)

- Reference architecture library
- Cost-benefit analysis / investment case templates
- Cross-engagement portfolio management (resource allocation, budget tracking, dependency mapping) — `/ea-status` provides a read-only portfolio dashboard only
- Architecture pattern detection / pattern library
- Multi-user collaboration with locking or conflict resolution
- External compliance framework mapping (GDPR, ISO 27001, HIPAA)
- SharePoint / cloud storage integration for requirements repository *(planned for a future version)*

This document is maintained in `ea-assistant/docs/PRD.md`. Update it when features are added or changed.